

Credit Risk Prediction and Explainable Credit Scoring System

Business Problem

Financial institutions incur significant losses due to loan defaults. The objective of this project is to develop a predictive framework capable of identifying customers likely to become seriously delinquent within the next two years.

Early identification of high-risk borrowers enables lenders to:

- Reduce credit losses
- Improve portfolio quality
- Support risk-based pricing
- Enhance underwriting decisions

Dataset Overview

Dataset: Give Me Some Credit

Records: 150,000 customers

Target Variable:

SeriousDlqin2yrs

- 1 = Default
- 0 = Non-default

Key Variables:

- Debt Ratio
- Monthly Income
- Revolving Credit Utilization
- Number of Late Payments
- Open Credit Lines
- Real Estate Loans

Methodology

The following workflow was implemented:

1. Data Understanding
2. Data Quality Assessment
3. Exploratory Data Analysis
4. Missing Value Treatment
5. Feature Engineering
6. Model Development
7. Model Evaluation
8. Explainable AI

Features engineered included:

- MonthlyDebt
- IncomePerDependent
- LatePaymentScore
- TotalPastDueEvents

Model Performance

Model	ROC-AUC
Logistic Regression	0.716
Random Forest	0.850
XGBoost	0.869

XGBoost achieved the strongest predictive performance and was selected as the preferred model.

Key Findings

- Historical delinquency was the strongest predictor of future default.
- Customers with repeated late payments exhibited significantly elevated risk.
- High revolving credit utilization increased default probability.
- Debt-related variables contributed strongly to risk classification.

Recommendations

1. Deploy XGBoost for borrower screening.
2. Introduce risk alerts for customers with repeated delinquency patterns.
3. Utilize SHAP explainability in analyst review workflows.
4. Monitor model performance on a quarterly basis.
5. Re-train credit risk models as portfolio characteristics evolve.

Business Value

The solution supports proactive borrower risk identification and helps financial institutions make more informed lending decisions. Incorporating explainable machine learning improves transparency and trust while maintaining strong predictive performance.